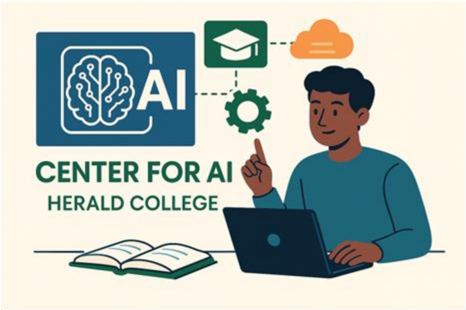
HCAI5TML01 – Mathematics of Learning.

Theme: Getting Started with Learning Theory.

Lec01 – Introduction to Statistical Learning Framework.

Is Learning Feasible?

Siman Giri



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1. Machine vs. Statistical Learning.

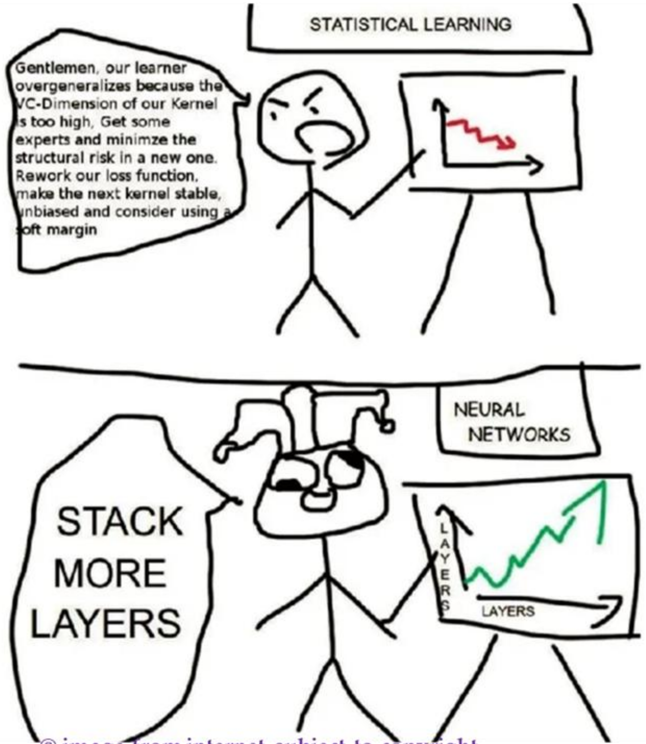
{Why do you think models with 100% training accuracy still fail in real-world tasks?}

1.1 Why Theorize Learning?

- **Machine learning works surprisingly well in practice** — but **why**?
  - **Because:**
    - Not all algorithms that perform well on training data succeed on unseen data.
      - **What governs this generalization?**
    - Some models memorize; others generalize.
      - **Can we predict this behavior mathematically?**
  - **Is successful learning always possible, or are there fundamental limits?**
    - Lead to **Statistical Learning Theory (SLT):**
      - A **framework** to ask, **when can learning succeed, and why?**

1.2 Machine Learning Vs. Statistical Learning.

- Machine Learning **asks how to build** models from data.
- **Statistical Learning Theory** **asks whether we can**, and **how well** we can expect those models to generalize.
- **SLT is to ML** what Physics is to Engineering: it **doesn't build the bridge**, but **it tells you if it will collapse**.



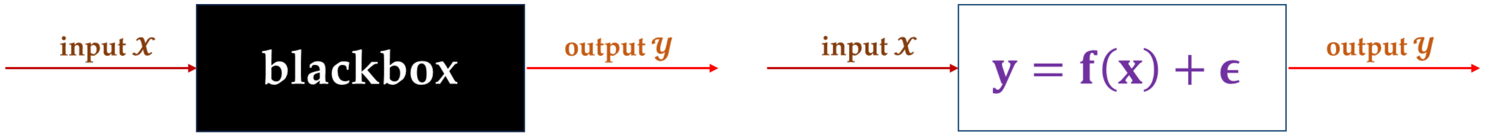


Fig: 1(a).

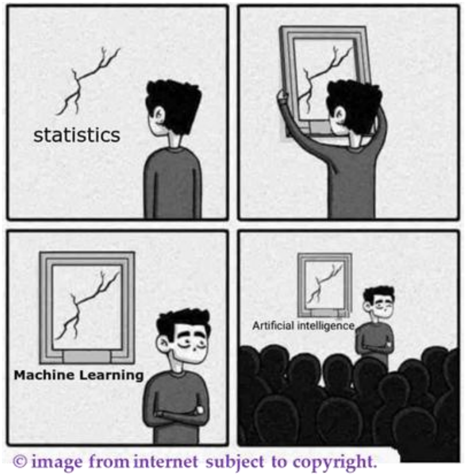
Fig: 1(b).

Fig 1: The ML black box (left) replicates input-output patterns from data, while the statistical approach (right) models the underlying relationships.

But ....

1.2.1 But ....

- The **traditional gap** between statistics and Machine Learning has **narrowed over the past few decades**.
  - Statistics increasingly addresses finite – sample behavior, model misspecification, and computational issues.
  - Machine learning has embraced probabilistic modeling, regularization, and theory – informed design.
- At their intersection lies **Statistical Learning Theory (SLT):**
  - a **field primarily concerned** with
    - **sample complexity**, **generalization guarantees**, and **the fundamental limits of learning**.



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- **SLT serves as a bridge**: it uses statistical tools to answer **ML questions** like:
  1. **"How much data do I need to learn?"**
     - *When is learning guaranteed to work?*
  2. **"How will my model behave on training?"**
     - *How complex can my model be before it overfits?*
  3. **"How will my model behave on unseen data?"**
     - *Can we bound the error on future data?*

1.3 Some Notations ….

- Let's **begin by reviewing the basic notation**:
  - We denote the **input vector as** $\mathbf{x} \in \mathbb{R}^{\mathbf{d}}$.
  - There exists an **unknown tar get function**: $\mathbf{f}\colon \boldsymbol{\mathcal{X}} \rightarrow \boldsymbol{\mathcal{Y}}$,
    - which **maps each input** $\mathbf{x}$ to a **corresponding label** $\mathbf{y} = \mathbf{f}(\mathbf{x})$.
  - **Here:**
    - $\mathcal{X}$ is the input space, containing all possible input vectors
    - $\mathcal{Y}$ is the output space, containing all possible labels or responses

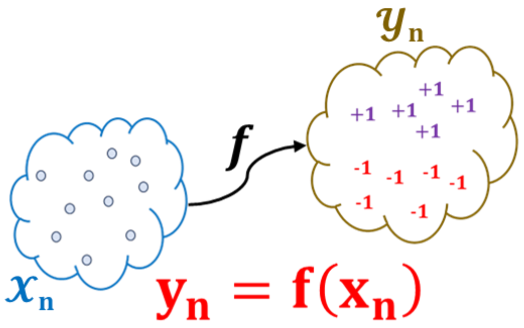


Fig: Input and Output in a Learning Framework.

1.4 Components of Learning Framework.

- Key Components:
  - *Input space* $\mathcal{X}$ *and Output space* $\mathcal{Y}$
  - *Unknown distribution* $\mathcal{D}$ *over* $\mathcal{X} \times \mathcal{Y}$
  - *Training data*: $\mathcal{S} = \{(x_1, y_1), \ldots, (x_n, y_n)\} \sim \mathcal{D}^n$ **[assuming supervised learning setup]**
  - *Hypothesis space* $\mathcal{H}$ **(set of functions:** $\mathcal{H} \subseteq \mathcal{Y} \wedge \mathcal{X}$**)**
  - *Loss function* $\ell(h(x), y)$ **(e.g 0 − 1 loss or squared loss)**

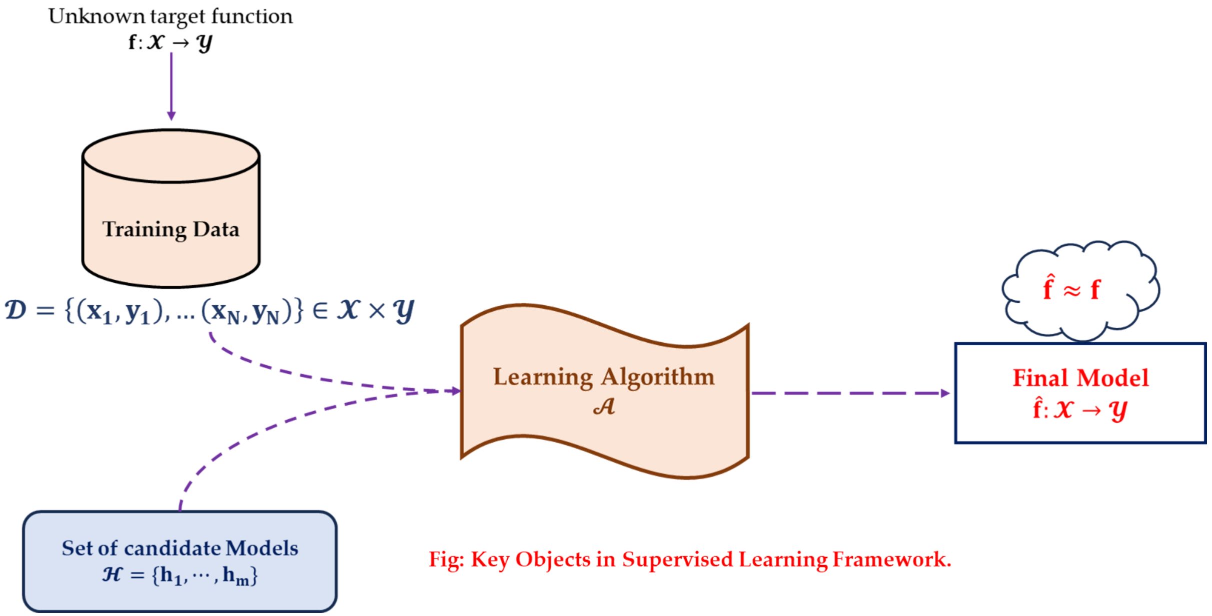


Fig: Key Objects in Supervised Learning Framework.

1.5 Supervised Learning Setup.

- We first assume the existence of a **target function f(.)**:
  - **target function** $\mathbf{f}: \mathcal{X} \rightarrow \mathcal{Y}$
- which is a function mapping from the **input's space** $\mathcal{X}$ to the **output space** $\mathcal{Y}$.
  - {Keep in mind, this **target function is idealized** and remains **unknown throughout learning**.}

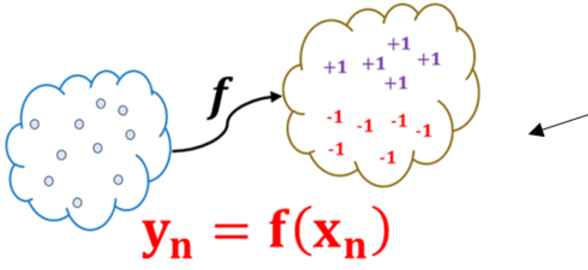


Fig: Input and Output in a Learning Framework.

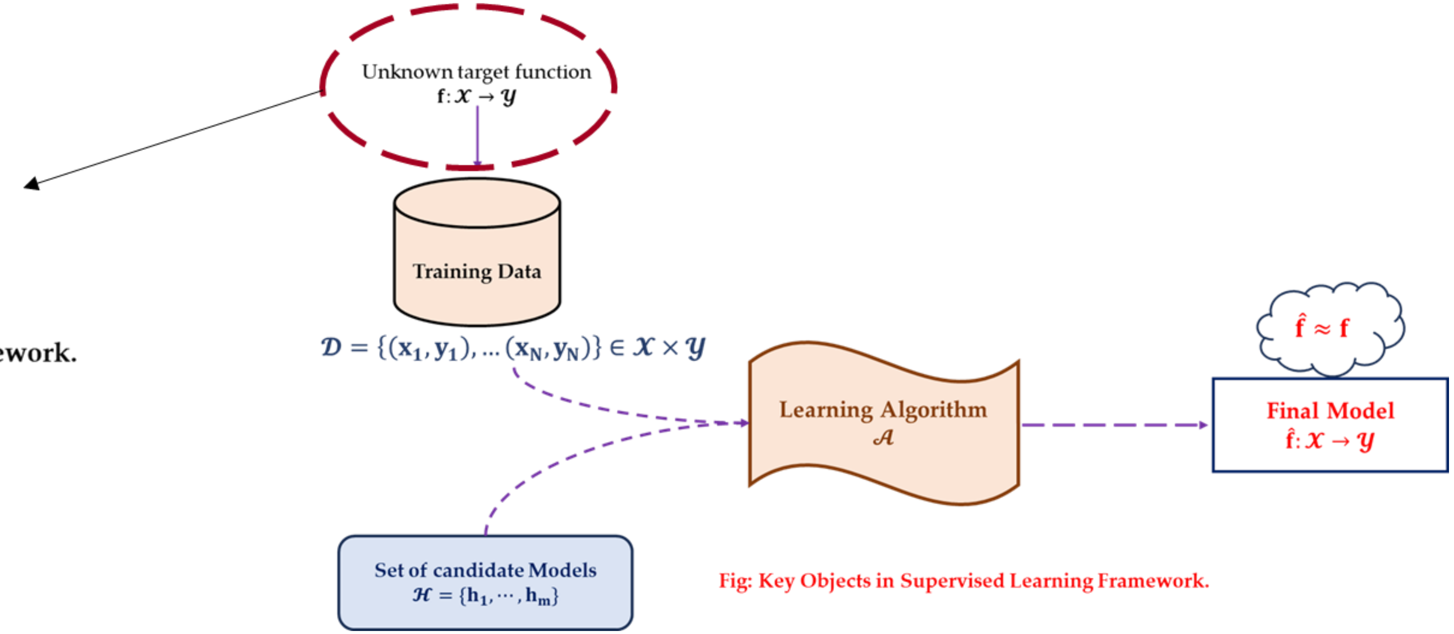
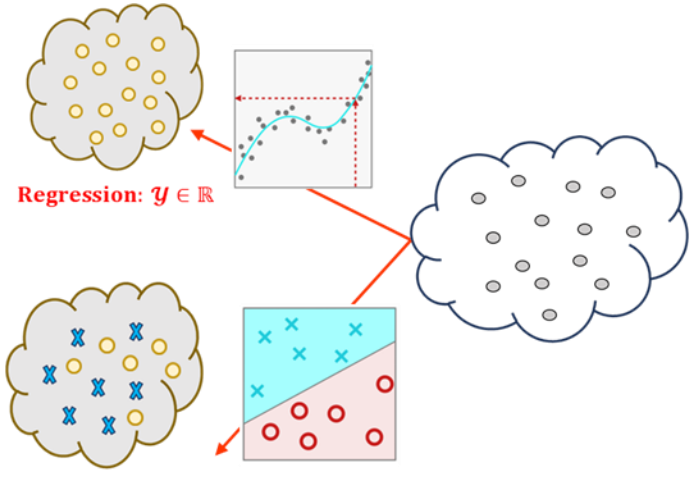


Fig: Key Objects in Supervised Learning Framework.

1.5.1 Supervised Learning Setup.

- Then, we have a **dataset $\mathcal{D}$:**
  - **dataset** $\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$
- where x_i **represents the vector of features** and y_i **represents the label for the i^{th} element.**



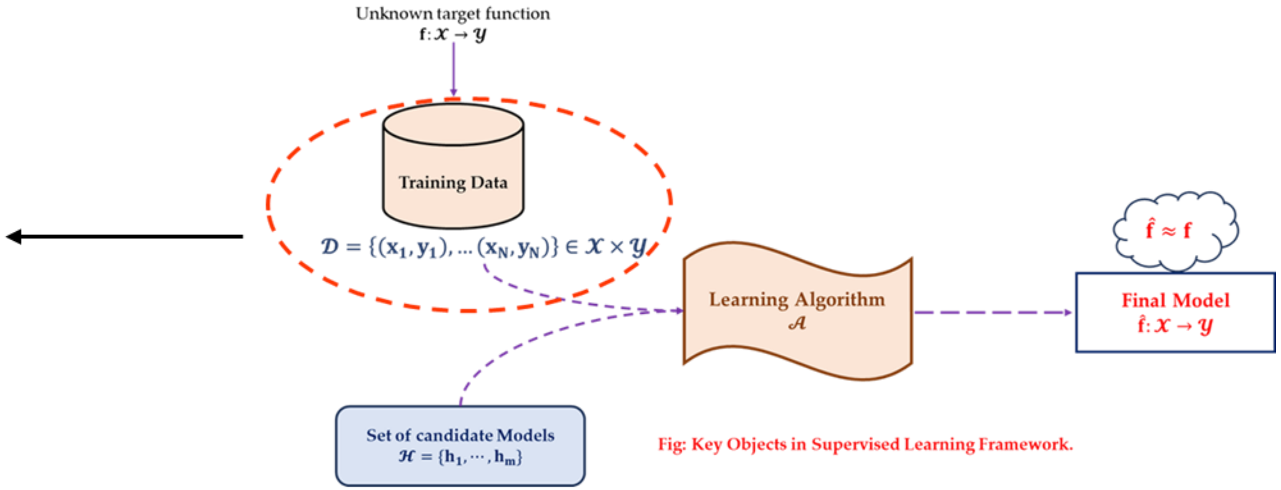


Fig: Key Objects in Supervised Learning Framework.

1.5.2 Supervised Learning Setup.

- From this data, we wish to obtain a fitted model, formally known as a **hypothesis model $\hat{f}(.)$:**
  - hypothesis model $\hat{f}: \mathcal{X} \rightarrow \mathcal{Y}$
- and then use **$\hat{f}$ to approximate** the **unknown target function f**.
- *Q: How can we* **find $\hat{f}$** *that* **best approximate** *the unknown* **target function f**?

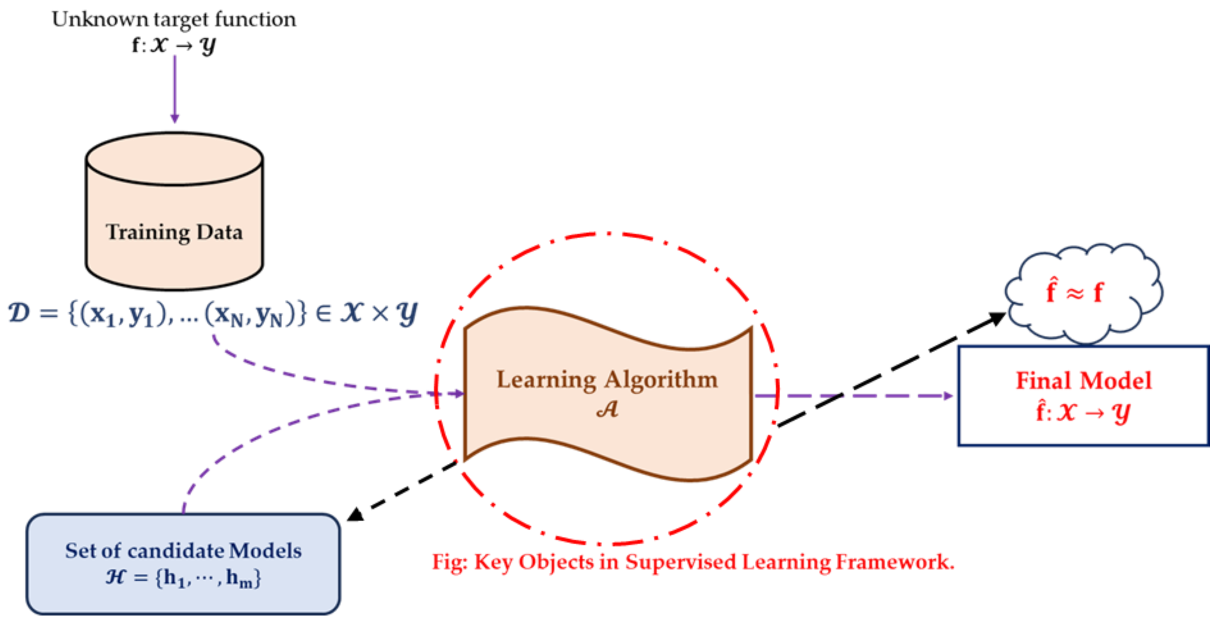


Fig: Key Objects in Supervised Learning Framework.

1.6 Why Learn from Data?.

- The **function f** is **unknown** and possibly too complex (**or even non – deterministic**).
- Analytical derivation of **f** or its **best approximation** would require full
  - knowledge of **how inputs map to outputs – which we do not have**.
- But we do have access to **examples called training datasets:**
  - **input output pairs** drawn from **some data generating process**.
- This leads us to the data driven paradigm:
  - Learn from the **past observation (data)** to **build a model** that **generalizes.**
  - This is the **essence of statistical learning**.

1.7 Hypothesis and Learning Algorithm?.

- In order **to find $\hat{\mathbf{f}}(.)$**, we typically consider a **set of candidate models**, also known as a **hypothesis set**,
  - $\mathcal{H} = \{\mathbf{h_1}, \mathbf{h_2}, \dots, \mathbf{h_m}\}$.
    - Each $\mathbf{h_1}, \cdots, \mathbf{h_m}$ are the candidate model (classifier or regressor) you want to evaluate.
    - The selected hypothesized model also called **hypothesis function $\mathbf{h_m^*}$**
      - will be the one used as our final model $\hat{\mathbf{f}}(.)$
- **How do we find $\hat{\mathbf{f}}(.)$?**
  - A **learning algorithm** searches for **the best hypothesis function h** $\in \mathcal{H}$ based on **the training data**.
- The next question is: What **makes selected function $\mathbf{h_m^*}$ "good"?**
  - In other words,
    - **How** do we evaluate how well $\mathbf{h_m^*}$ **approximates** the true (but unknown) **target function f?**
  - Since **f is not observable**, we cannot directly compare $\mathbf{h_m^*}$ to **f.**
  - Instead, we assess the quality of $\mathbf{h_m^*}$ by measuring its **expected error (or risk)** over the **data distribution.**
    - which tells us **how closely h mimics f on unseen examples**.
- **This leads to Empirical Risk Minimization Framework.**

2. Getting Started with Learning.

{Empirical Risk Minimization Framework.}

2.1 Idea of Good Predictions …

- What is the **ultimate goal** in "**supervised learning**"?
  - Quick answer, we want a "**good**" model.
  - What does "**good**" model mean?
    - Simply put, it means that we want to **estimate an unknown model f** with **some model $\hat{f}$**
      - that gives "**good**" **predictions**.
  - What do we mean by "**good**" **predictions**?
    - Loosely speaking, it means that we want to obtain "**accurate**" **predictions**.
      - notion of *accurate predictions*, **minimum expected loss (or risk) aka Error.**
- **Two types of Predictions:**
  - we deal with two corresponding types of predictions:
    - **Predictions $\hat{y}_i$ of observed/seen values x_i**
    - **Predictions $\hat{y}_o$ of unobserved/unseen values x_o**
  - From the supervised learning standpoint,
    - we are ultimately *interested in **predictions** $\hat{y}_o$*
      - and we want this to be as **accurate as possible**.

2.2 Two Types of Data.

- Two distinct types of predictions involve two slightly different kinds of data.
- **To summarize:**
  - **In – sample data:**
    - These are **the data points** that live inside an **example dataset also called training dataset**.
    - Denoted $\mathcal{D}_{\text{in}}$ and are **used to fit a model.**
  - **Out – sample data:**
    - These are **data points** that **are not present in a training set** $\mathcal{D}_{\text{in}}$ also called **testing data**.
    - Denoted $\mathcal{D}_{\text{out}}$ and are used to **measure the predictive quality of a model**.

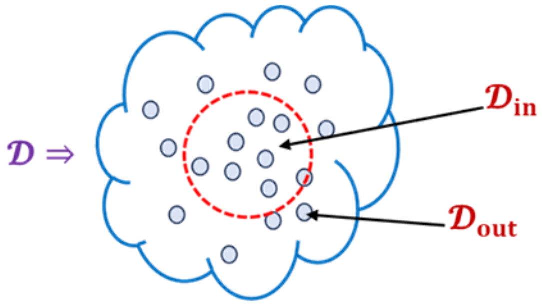
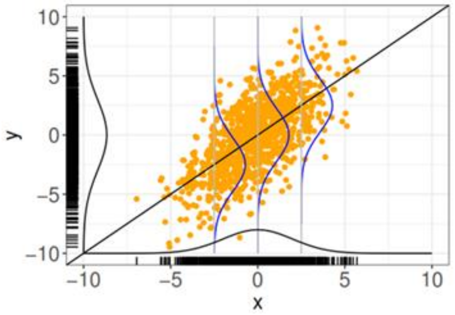


Fig: In and Out Sample Data.

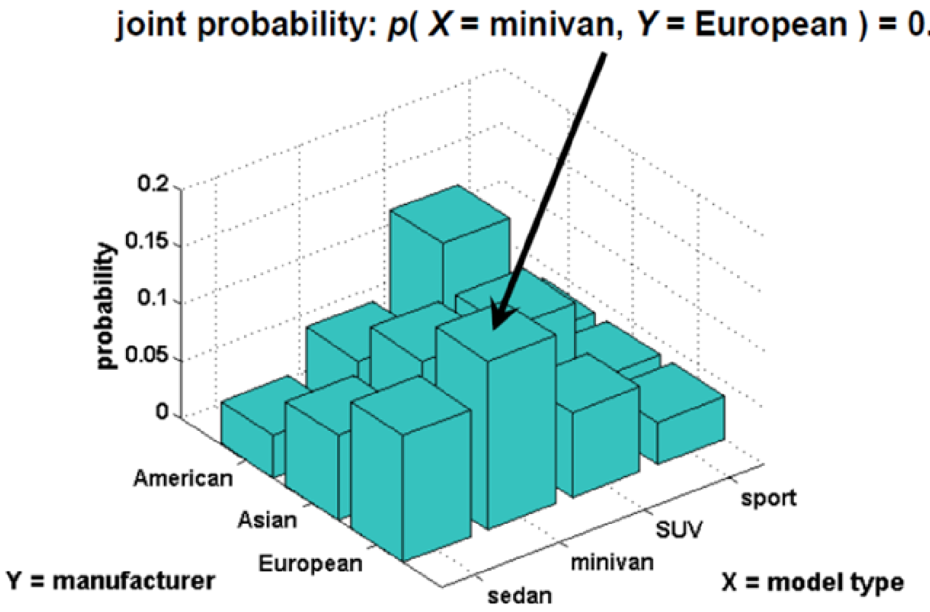
2.3 Data: Assumptions about Data Generating Process.

- We assume the **observed data to be generated by a process**
  - that can be **characterized by some probability distribution**:
    - $\mathbb{P}_{x,y}$ defined on $\mathcal{X} \times \mathcal{Y}$.
  - It is **important to understand** that the **true distribution** is **essentially unknown** to us.
  - In a certain sense, learning (part of) its structure is what ML is all about.
- We assume data to be drawn **i.i.d**
  - from the **joint probability density function (pdf)/probability mass function (pmf)** → $\mathbf{p(x, y)}$.
  - **i.i.d** stands for *independent and identically distributed*.
  - This means:
    - we assume that all samples are drawn from **the same distribution and are mutually independent**.
    - the $\mathbf{i^{th}}$ **realization does not depend** on the **other** $\mathbf{i-1^{th}}$ **or** $\mathbf{i+1^{th}}$ **realizations**.
- This is a **strong yet crucial assumption** that is precondition to most theory in (basic) ML.



Joint Probability Distribution.

- **Probability distribution** that acts on **many variables at the same time** is known as a
  - **joint probability distribution**.
- Given any ***values x and y of two random variables X and Y***,
  - what is the probability that ***X = x and Y = y simultaneously***?
    - ***P(X = x, Y = y)*** **denotes the joint probability** also **written p(x, y).**



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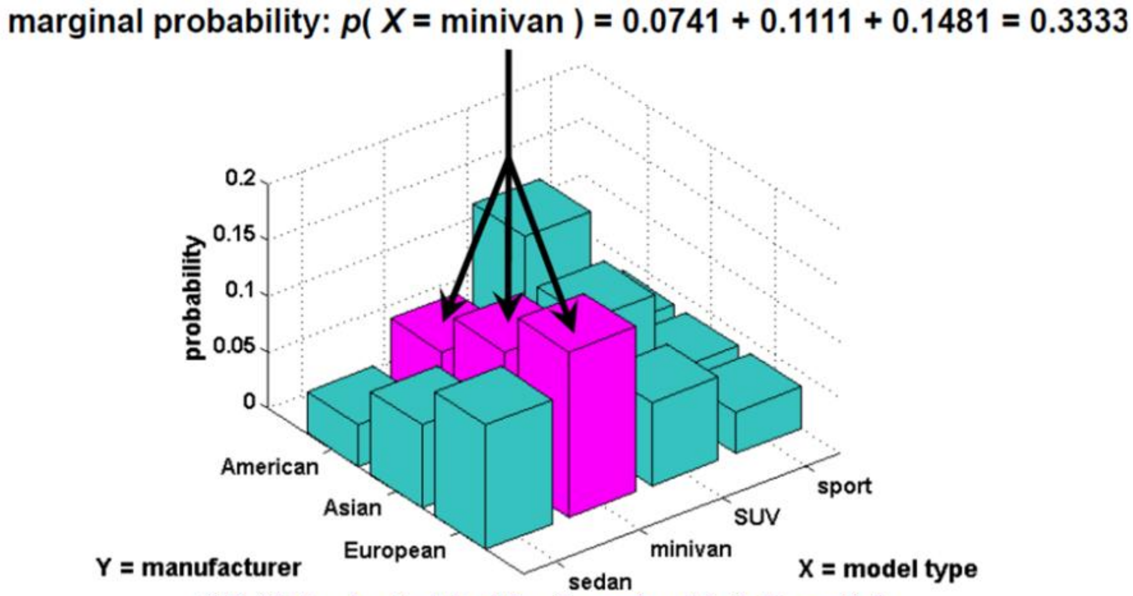
Marginal Probability Distribution.

- **Marginal probability distribution** is the **probability distribution** of a **single variable**.
- It is **calculated based** on the **joint probability distribution** $P(X, Y)$
  - I.e., using the sum rule:
    - $P(X = x) = \sum y \; P\;(X = x, Y) = y$
  - For continuous random variables, the summation is replaced with integration,
    - $\int \mathbf{P}\left(\mathbf{X} = \mathbf{x}, \mathbf{Y_y}\right)\mathbf{dy}$
- This process is called marginalization.



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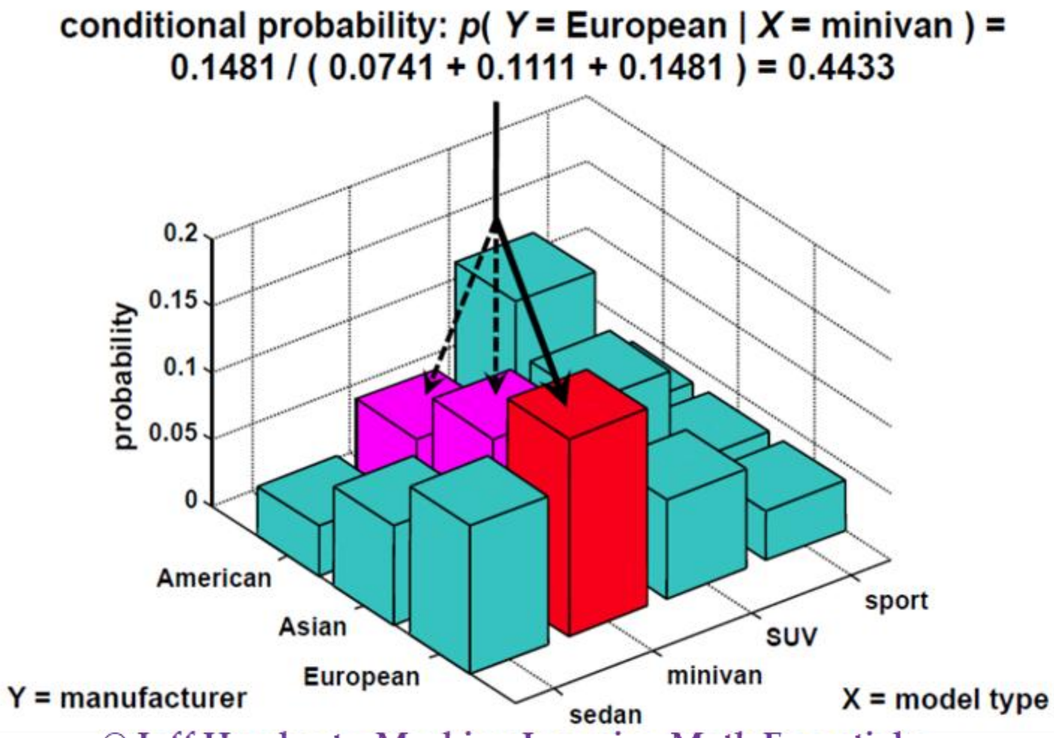
Conditional Probability Distribution.

- **Conditional probability distribution** is the **probability distribution of one variable**
  - provided that **another variable** has **taken a certain value**.
    - **Denoted** $\mathbf{P(X = x|Y = y)}$.



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Joint, Conditional and Marginal Distribution – Example.

Joint Distributions

Represents the probabilities of both weather and activity combinations:

P(Rainy and Walking)=0.2

P(Rainy and Running)=0.1

P(Sunny and Walking)=0.4

P(Sunny and Running)=0.3

	Walking	Running	Marginal - Weather
Rainy	0.2	0.1	0.3
Sunny	0.4	0.3	0.7
Marginal - Activity	0.6	0.4	1

Marginal Distributions

For Weather:

P(Rainy)=0.2+0.1=0.3

P(Sunny)=0.4+0.3=0.7

For Activity:

P(Walking)=0.2+0.4=0.6

P(Running)=0.1+0.3=0.4

Conditional Distribution:

For Activity given Weather:

$$P(\text{Walking}|\text{Rainy}) = \frac{P(\text{Rainy and Walking})}{P(\text{Rainy})} = \frac{0.2}{0.3} = 0.6667$$

$$P(\text{Running}|\text{Rainy}) = \frac{P(\text{Rainy and Running})}{P(\text{Rainy})} = \frac{0.1}{0.3} = 0.3333$$

For Weather given Activity:

$$P(\text{Rainy}|\text{Walking}) = \frac{P(\text{Rainy and Walking})}{P(\text{Walking})} = \frac{0.2}{0.6} = 0.6667$$

$$P(\text{Sunny}|\text{Walking}) = \frac{P(\text{Rainy and Walking})}{P(\text{Walking})} = \frac{0.4}{0.6} = 0.3333$$

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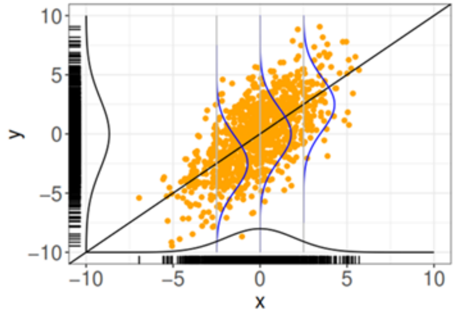


Big Q: What happens if we do not make such assumptions?

2.4 When is Learning not Feasible?

- Let's **imagine that training samples** are selected **deterministically**
  - a fixed **set of datapoints** $\{x_1, \ldots, x_N\}$ with **no randomness**.
- In **such a setting**,
  - where **both training and test data** come from **an arbitrary, unknown, and possibly unrelated process**,
  - there's **no reason to believe** that what **we learn from training data** will **generalize to unseen data**.
- Without a shared underlying **probability distribution** linking training and test data,
  - **learning becomes meaningless**.
- The probabilistic assumption is what connects the training data to the real world
  - **and makes learning possible.**
- E.g.:
  - Learning without a shared distribution is like practicing basketball on a 6-foot hoop and expecting to perform equally well on a regulation 10-foot hoop.

2.5.1 Statistical Learning Theory.

- Define a space where a decision or prediction function is applicable:
  - Assume there is a data generating distribution $\mathbb{P}_{\mathcal{X}\times\mathcal{Y}}$.
  - All input/output pairs (x, y) are generated i.i.d from $\mathbb{P}_{\mathcal{X}\times\mathcal{Y}}$.
  - One common expectation is to have prediction function $\mathbf{f(x)}$ "that does well on average" i.e. :
    - $\boldsymbol{\ell}(\mathbf{f(x), y})$ is usually small.

Risk – Definition:
The **risk** of a prediction function $\mathbf{f}\text{: } \mathcal{X} \to \mathcal{A}$ is:

$$\mathbf{R(f)} = \mathbb{E}_{(\mathbf{x,y})\sim\mathbb{P}(\mathcal{X}\times\mathcal{Y})}[\boldsymbol{\ell}(\mathbf{f(x), y})]$$

In words, it's the expected loss of **f over** $\mathbb{P}_{\mathcal{X}\times\mathcal{Y}}$.

- We can not compute the risk function, because we do not know the theoretical distribution or **true** $\mathbb{P}_{\mathcal{X}\times\mathcal{Y}}$ hence expectation could not be computed.
  - But in **Machine Learning**/statistics or data science we can **estimate** it based on our empirical observation or sampled data.

2.5.2 Empirical Risk.

- Empirical risk is an estimated risk of theoretical risk, and computed on provided set of sample data.
- Thus for any sample data:
  - Let $\mathcal{D}_n = \{(x_1, y_1), \dots, (x_n, y_n)\}$ be drawn **i.i.d** from $\mathbb{P}_{\mathcal{X}\times\mathcal{Y}}$.
- Empirical Risk Could be defined as:

Empirical Risk – Definition:
The **empirical risk** of a prediction function $\mathbf{f}: \mathcal{X} \rightarrow \mathcal{A}$ with respect to $\mathcal{D}_n$ is:

$$\hat{R}_n(f) = \frac{1}{n}\sum_{i=1}^{n} \ell(f(x_i), y_i)$$

Backed by strong law of large numbers, which gives:
$$\lim_{n\to\infty} \hat{R}_n(f) = R(f)$$

- Thus the empirical risk is average loss on our sampled data, we expect it to be as minimum as possible,

2.5.2 Empirical Risk.

- Thus we want function **f** described by set of parameters with $\boldsymbol{\theta}^* \in \boldsymbol{\Theta}$ which produces the **minimum loss** on **average**.
- So Remember the question:
  - How do we fit or train the model so we get best set of parameters?
  - How do we determined which set of parameters best fits our function or model?

2.5.2 Empirical Risk.

- Thus we want function **f** described by set of parameters with $\boldsymbol{\theta}^* \in \boldsymbol{\Theta}$ which produces the **minimum loss** on **average**.
- So Remember the question:
  - How do we fit or train the model so we get best set of parameters?
    - Using evaluation measure also called as loss function, which in general is the difference between true label and predicted label.
  - How do we determined which set of parameters best fits our function or model?
    - Which ever set of parameters describes the function and produce the the minimum risk(estimated), which is an average loss on all our sampled data points.
- **The answer could be collectively described by → Empirical Risk Minimization.**

2.5.3 Empirical Risk Minimization.

- Among a collection of candidate function $\mathbf{f}$
  - described by set of parameters $\boldsymbol{\theta} \in \boldsymbol{\Theta}$
  - we desired to select a function $\mathbf{f}$ learned from sample data $\mathcal{D}_n\{(\mathbf{x_i}, \mathbf{y_i})\}$
  - which is described by parameter $\boldsymbol{\theta}^* \in \boldsymbol{\Theta}$ and
  - produces minimum average loss – Risk $\widehat{\mathbf{R}}(\mathbf{f})$.
- Formally:

Empirical Risk Minimizer – Definition:
A function $\hat{\mathbf{f}}$ is an empirical risk minimizer if:

$$\hat{\mathbf{f}} \in \mathbf{argmin}_{\mathbf{f}}\, \widehat{\mathbf{R}}_{\mathbf{n}}(\mathbf{f}),$$

where the minimum is taken over all functions $\mathbf{f}\colon \mathcal{X} \to \mathcal{A}$.

3. Risk to Error Measures.

3.1 In – sample Error.

- From a **dataset $\mathcal{D} = \{\mathbf{x_1}, \dots \mathbf{x_N}\}$** and **set of hypothesis $\mathcal{H} = \{\mathbf{h_1}, \dots \mathbf{h_m}\}$**
  - Suppose that we pick a hypothesis h that approximates the function f.
- For every in – sample x_n,
  - we check whether **the output returned by h** is **the same as** **the output returned by f** i.e.
    - $\{\mathbf{h(x_n) == f(x_n)}\}$, **for $\mathbf{n = 1, \dots, N}$.**
  - If $\{\mathbf{h(x_n) == f(x_n)}\}$, then we say that the **in – sample $\mathbf{x_n}$** is **correctly classified** in **the training**.
  - If $\{\mathbf{h(x_n) \neq f(x_n)}\}$, then we say that $\mathbf{x_n}$ **is incorrectly classified**.
- Averaging over all N samples, we obtain a quantity called the **in – sample error or the training error**.

Definition 1 In – sample Error:

- Consider a training set $\mathcal{D} = \{\mathbf{x_1}, \dots \mathbf{x_N}\}$, and a target **function f**. The in – sample error (or the training error) of a hypothesis function $\mathbf{h} \in \mathcal{H}$ is the empirical average of $\{\mathbf{h(x_n) \neq f(x_n)}\}$
  - $\mathbb{E}_{in}(\boldsymbol{h}) \overline{\overline{def}} \frac{1}{N}\sum_{n=1}^{N}[\![h(x_n) \neq f(x_n)]\!]$,
  - where $[\![.]\!] = \mathbf{1}$ if the statement inside the bracket is true, and = 0 if the statement is false.

3.2 Out – sample Error.

- How about the out – samples?
  - Since we assume that x is drawn from a distribution $\mathbf{p}_{\mathcal{X}}(\mathbf{x})$, we can define the out – sample error
    - as the **probability** that $\{\mathbf{h}(\mathbf{x}) \neq \mathbf{f}(\mathbf{x})\}$ for all $\mathbf{x} \sim \mathbf{p}_{\mathcal{X}}(\mathbf{x})$.

Definition 2 Out – sample Error:
- Consider an **input space $\mathcal{X}$** containing elements **x drawn from a distribution $\mathrm{p}_{\mathcal{X}}(\mathrm{x})$**, and a **target function** f.
- The out – sample error (or testing error) of a hypothesis function **h** $\in \mathcal{H}$ is:
  - $\mathbb{E}_{out}(h) \overline{\overline{def}}\ \mathbb{P}[h(x) \neq f(x)]$;
  - where $\mathbb{P}[.]$ measures **the probability of the statement based on the distribution $p_{\mathcal{X}}(x)$.**

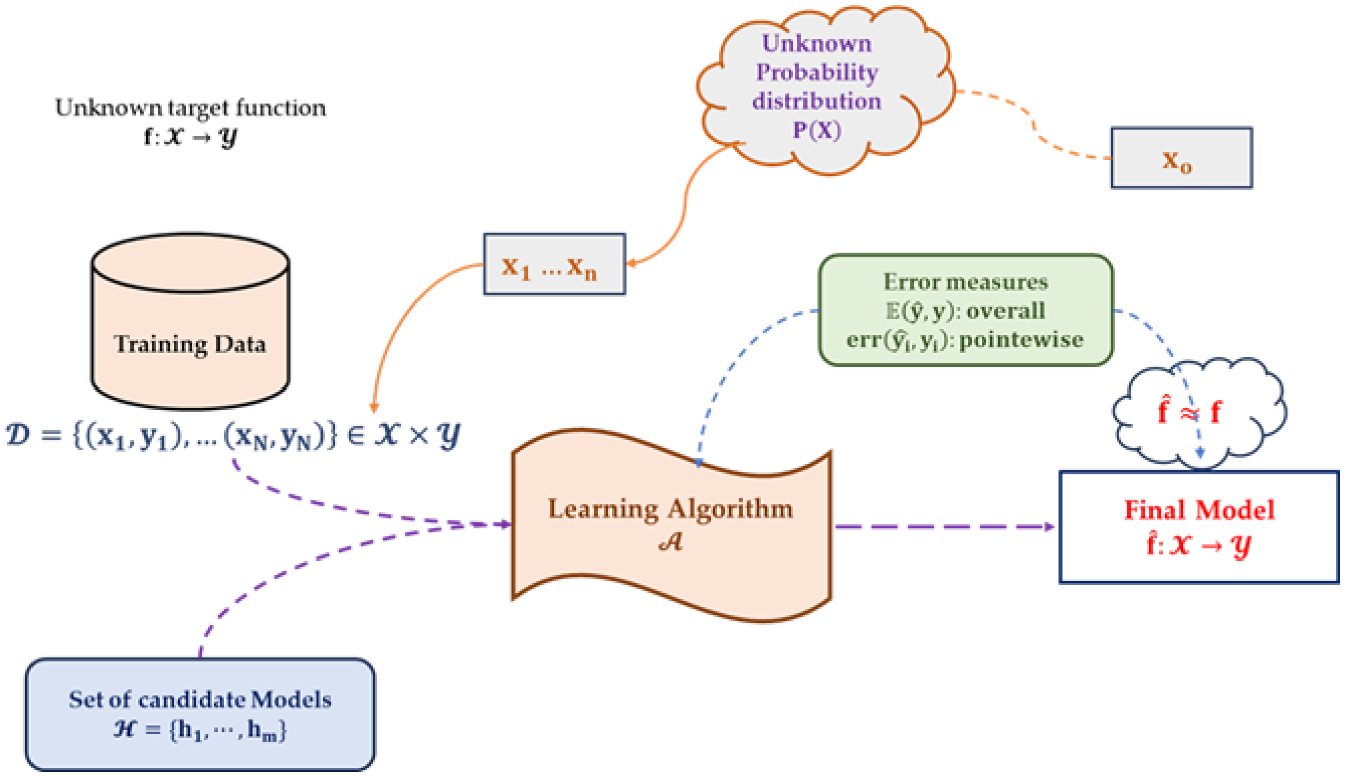
- Since $[\![.]\!]$ is a binary function, the **out – sample error** is the **expected value of sample being misclassified**
- **over the entire distribution:**

$$\begin{aligned}\mathbb{E}_{\mathbf{out}}(\mathbf{h}) &= \mathbb{P}[\mathbf{h}(\mathbf{x}) \neq \mathbf{f}(\mathbf{x})] \\ &= [\![\mathbf{h}(\mathbf{x_n}) \neq \mathbf{f}(\mathbf{x_n})]\!]\mathbb{P}\{\mathbf{h}(\mathbf{x_n}) \neq \mathbf{f}(\mathbf{x_n})\} + [\![\mathbf{h}(\mathbf{x_n}) = \mathbf{f}(\mathbf{x_n})]\!](\mathbf{1} - \mathbb{P}\{\mathbf{h}(\mathbf{x_n}) \neq \mathbf{f}(\mathbf{x_n})\}) \\ &= \mathbb{E}\{[\![\mathbf{h}(\mathbf{x_n}) \neq \mathbf{f}(\mathbf{x_n})]\!]\}\end{aligned}$$

3.3 Why this Matter?

- The relationship between **in-sample error $\mathbb{E}_{in}(h)$** and out-of-sample error $\mathbb{E}_{out}(h)$ mirrors the relationship between:
  - **"the empirical average and"**
  - **"the true mean of a random variable".**
    - $\mathbb{E}_{in}(h)$: average loss on the training data (empirical estimate)
    - $\mathbb{E}_{out}(h)$: expected loss on unseen data (true/generalization error)
- **Why this matters**:
  - It justifies **statistical learning:**
    - **we use data (sample) to estimate how well we might perform in the real world (population).**

Final Learning Framework.



Thank You